1. Lesson 8

1.1. Exercise + Challenge:

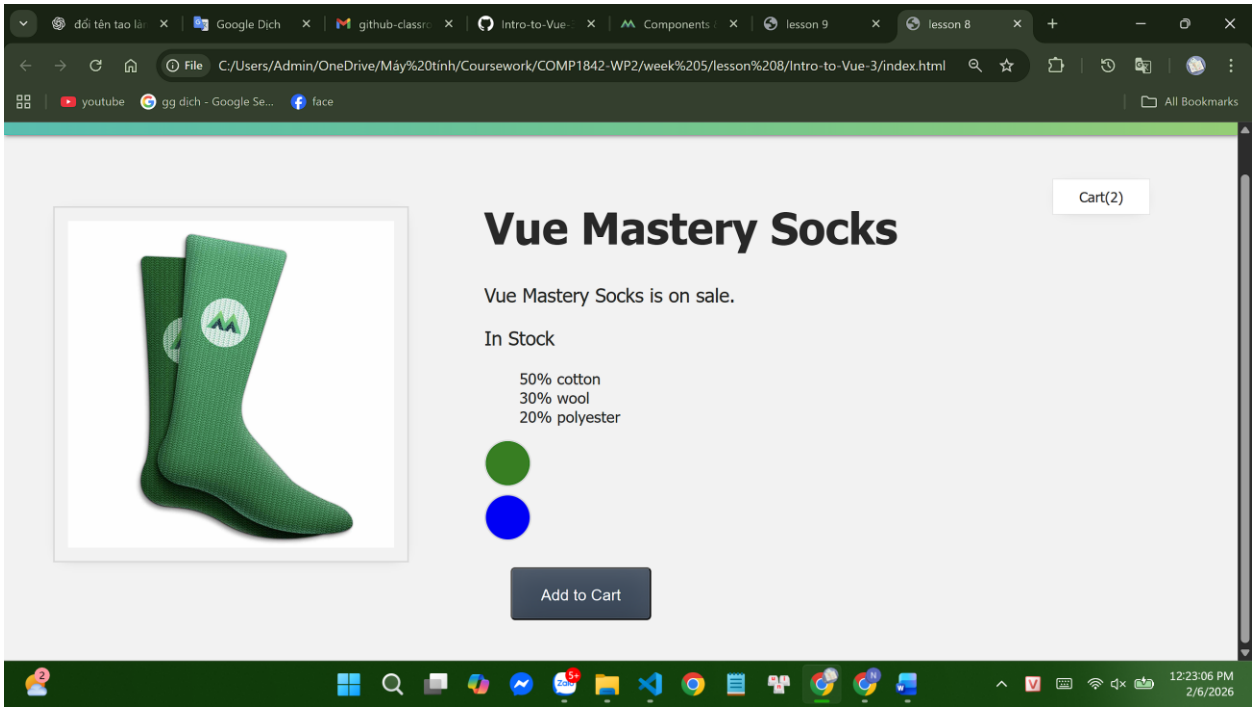
```html
<html lang="en">
  <body>
    <div id="app">
      <div class="product-display">
        <div class="product-container">
          <div class="product-info">
            <!--Exercise--> <!--Ngo Quoc Cuong-->
            <h1>{{ title }}</h1>
            <!--Exercise-->

            <!--Challenge-->
            <p>{{ sale }}</p>
            <!--Challenge-->
            <p v-if="inStock">In Stock</p>
            <p v-else>Out of Stock</p>
            <ul>
              <li v-for="detail in details">{{ detail }}</li>
            </ul>

            <!--exercise-->
            <div
              v-for="(variant, index) in variants"
              :key="variant.id"
              @mouseover="updateVariant(index)"
              class="color-circle"
              :style="{ backgroundColor: variant.color }">
            </div>
            <!--exercise-->
```

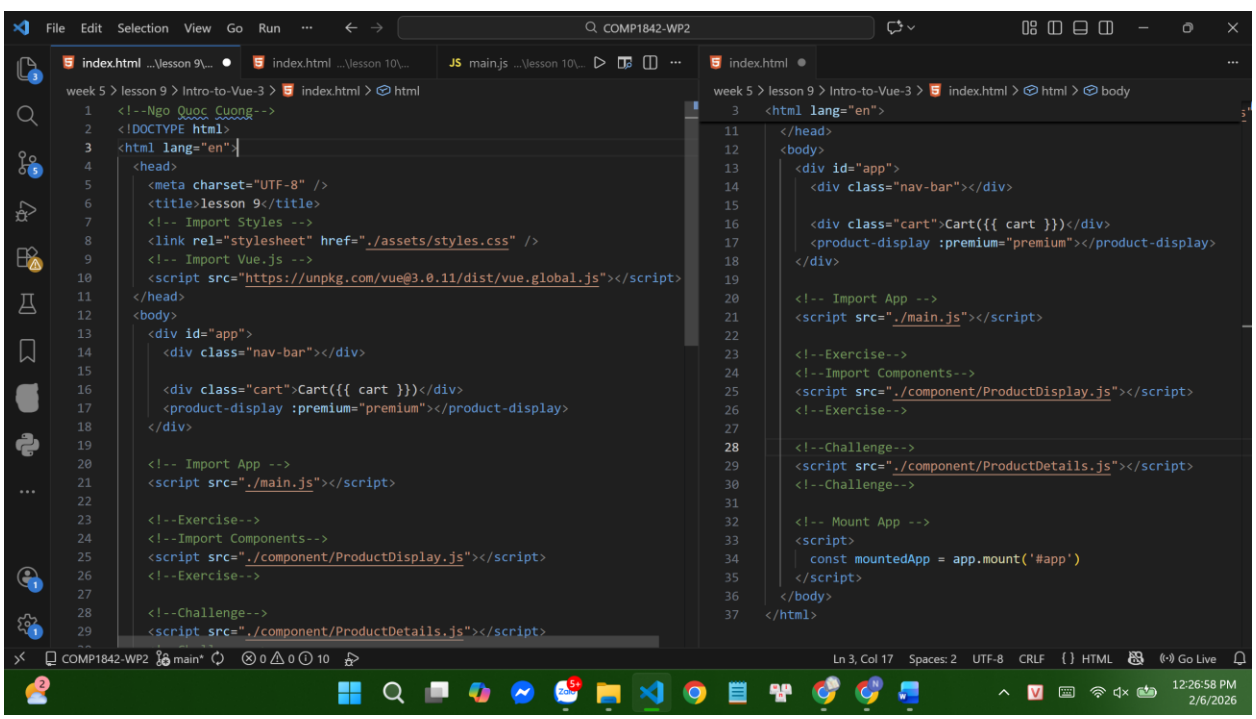
```javascript
//Ngo Quoc Cuong
const app = Vue.createApp({
    data() {
        return {
            cart:0,
            product: 'Socks',
            brand: 'Vue Mastery',
            selectedVariant: 0,
            details: ['50% cotton', '30% wool', '20% polyester'],
            variants: [
              { id: 2234, color: 'green',
                image: './assets/images/socks_green.jpg', quantity: 50 }
              { id: 2235, color: 'blue', image:
                './assets/images/socks_blue.jpg', quantity: 0 },
            ],
            //challenge
            onSale: true
            //challenge
        }
    },
    methods: {
        addToCart() {
            this.cart += 1
        },
        //exercise
        updateVariant(index) {
            this.selectedVariant = index
        }
        //exercise
```

```javascript
const app = Vue.createApp({
    methods: {
    },
    //exercise
    computed: {
        title() {
            return this.brand + ' ' + this.product
        },
        image() {
            return this.variants[this.selectedVariant].image
        },
        inStock() {
            return this.variants[this.selectedVariant].quantity
        },
    //exercise
        //challenge
        sale() {
            if (this.onSale) {
                return this.brand + ' ' + this.product
                + ' is on sale.'
            }
            return ''
        }
        //challenge
    }
})
```



2. Lesson 9

2.1.Exercise:



```javascript
//exercise
app.component('product-display', {
  props: {
    premium: {
      type: Boolean,
      required: true
    }
  },
  template:
  /*html*/
  `<div class="product-display">
    <div class="product-container">
      <div class="product-image">
        <img v-bind:src="image">
      </div>
      <div class="product-info">
        <h1>{{ title }}</h1>

        <p v-if="inStock">In Stock</p>
        <p v-else>Out of Stock</p>

        <p>Shipping: {{ shipping }}</p>

        <!--Challenge-->
        <product-details :details="details"></product-details>
        <!--Challenge-->

        <div
          v-for="(variant, index) in variants"
```

```html
    template:
          v-for="(variant, index) in variants"
          :key="variant.id"
          @mouseover="updateVariant(index)"
          class="color-circle"
          :style="{ backgroundColor: variant.color }">
        </div>

        <button
          class="button"
          :class="{ disabledButton: !inStock }"
          :disabled="!inStock"
          v-on:click="addToCart">
          Add to Cart
        </button>
      </div>
    </div>
  </div>`,
  data() {
    return {
      product: 'Socks',
      brand: 'Vue Mastery',
      selectedVariant: 0,
      details: ['50% cotton', '30% wool', '20% polyester'],
      variants: [
        { id: 2234, color: 'green', image: './assets/images/soc
        { id: 2235, color: 'blue', image: './assets/images/sock
      ]
    }
```

```javascript
  data() {
      ]
    }
  },
  methods: {
      addToCart() {
        this.cart += 1
      },
      updateVariant(index) {
        this.selectedVariant = index
      }
  },
  computed: {
      title() {
        return this.brand + ' ' + this.product
      },
      image() {
        return this.variants[this.selectedVariant].image
      },
      inStock() {
        return this.variants[this.selectedVariant].quantity
      },
      shipping() {
        if (this.premium) {
          return 'Free'
        }
        return 2.99
      }
  }
```

```javascript
  computed: {
      shipping() {
        return 2.99
      }
  }
})
//exercise Ngo Quoc Cuong
```

2.2.Challenge:

File Edit Selection View Go Run COMP1842-WP2

roductDisplay.js ...\component | ProductDisplay.js ...\Components | ProductDetails.js | main.js ...\lesson 8\... | index.html ...\lesson 7\... | main.js ...\lesson 7\...

week 5 > lesson 9 > Intro-to-Vue-3 > component > ProductDetails.js > ...

```
//challenge
app.component('product-details', {
  props: {
    details: {
      type: Array,
      required: true
    }
  },
  template:
  /*html*/
  `
  <ul>
    <li v-for="detail in details">{{ detail }}</li>
  </ul>
  `
})
//challenge Ngo Quoc Cuong
```

COMP1842-WP2 main* 0 0 14 Ln 17, Col 27 Spaces: 4 UTF-8 CRLF { } JavaScript Go Live

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File Edit Selection View Go Run COMP1842-WP2

main.js ...\lesson 9\... | index.html ...\lesson 9\... | index.html ...\lesson 10\... | main.js ...\lesson 10\... | ProductDisplay.js ...\component | ProductDisplay.js ...\C

week 5 > lesson 9 > Intro-to-Vue-3 > main.js > ...

```
const app = Vue.createApp({
    data() {
        return {
            cart: 0,
            premium: true
        }
    },
    methods: {}
})
// Ngo Quoc Cuong
```

COMP1842-WP2 main* 0 0 16 Ln 10, Col 18 Spaces: 4 UTF-8 CRLF { } JavaScript Go Live

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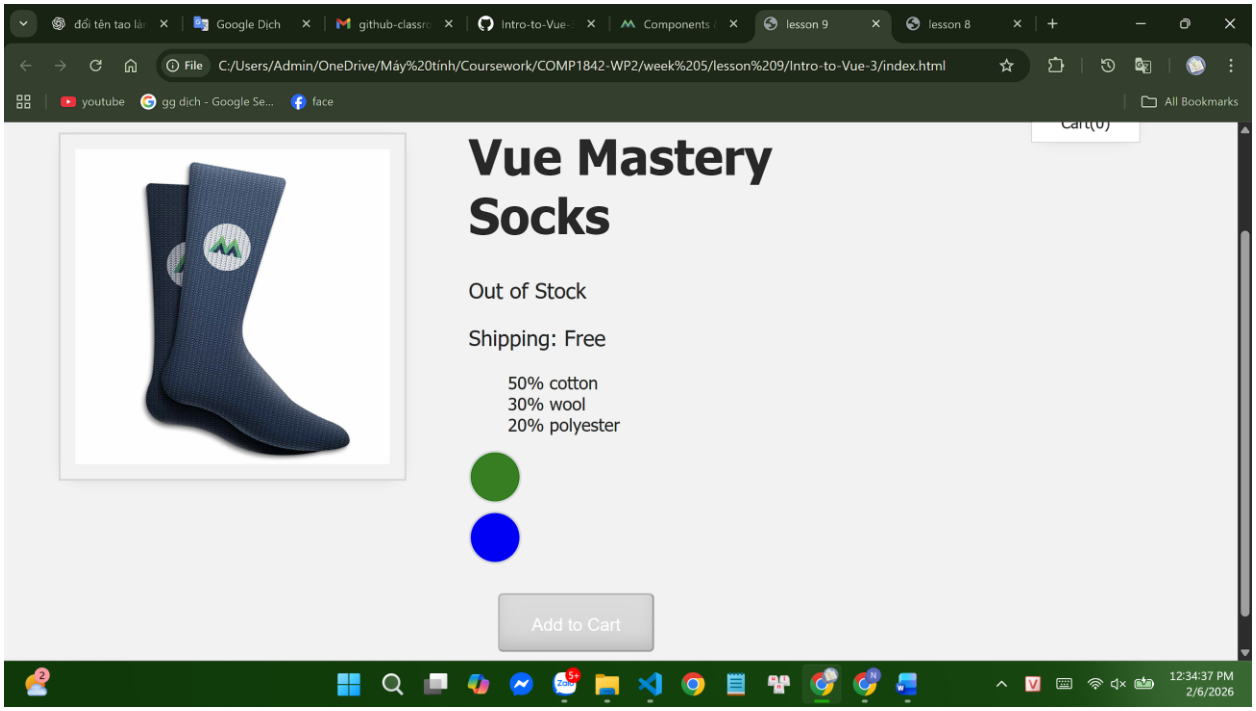
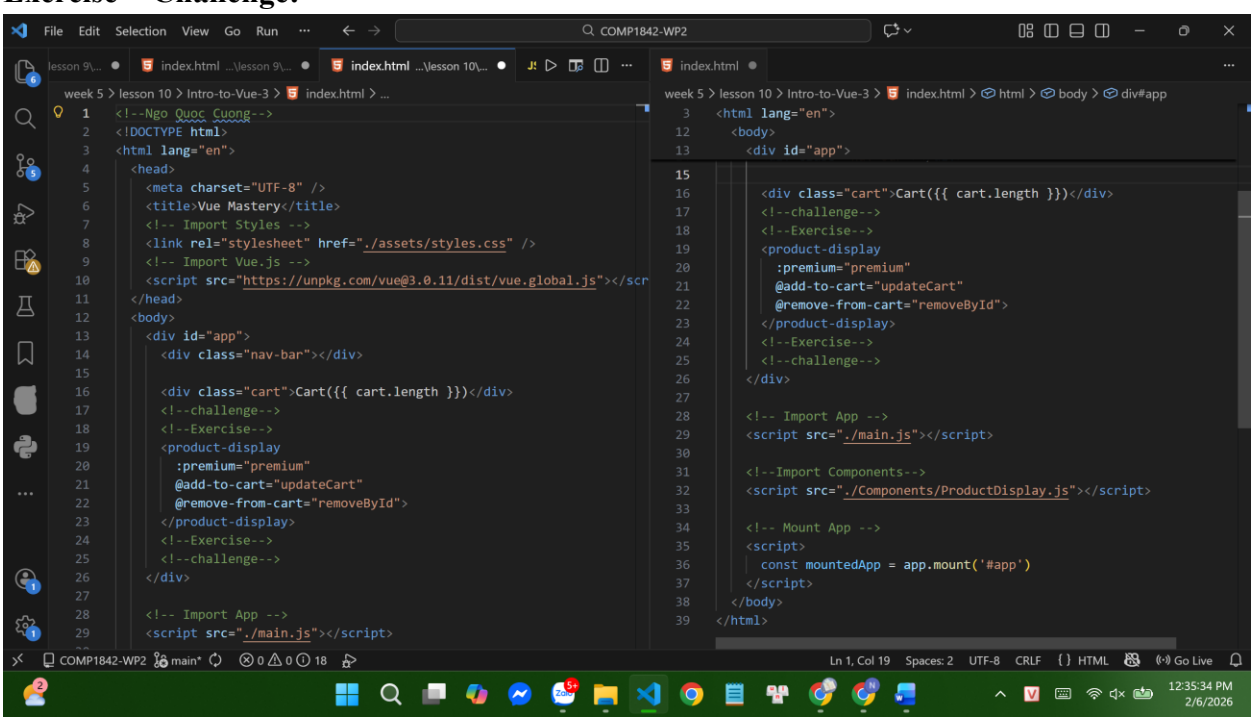


3. Lesson 10

Exercise + Challenge:



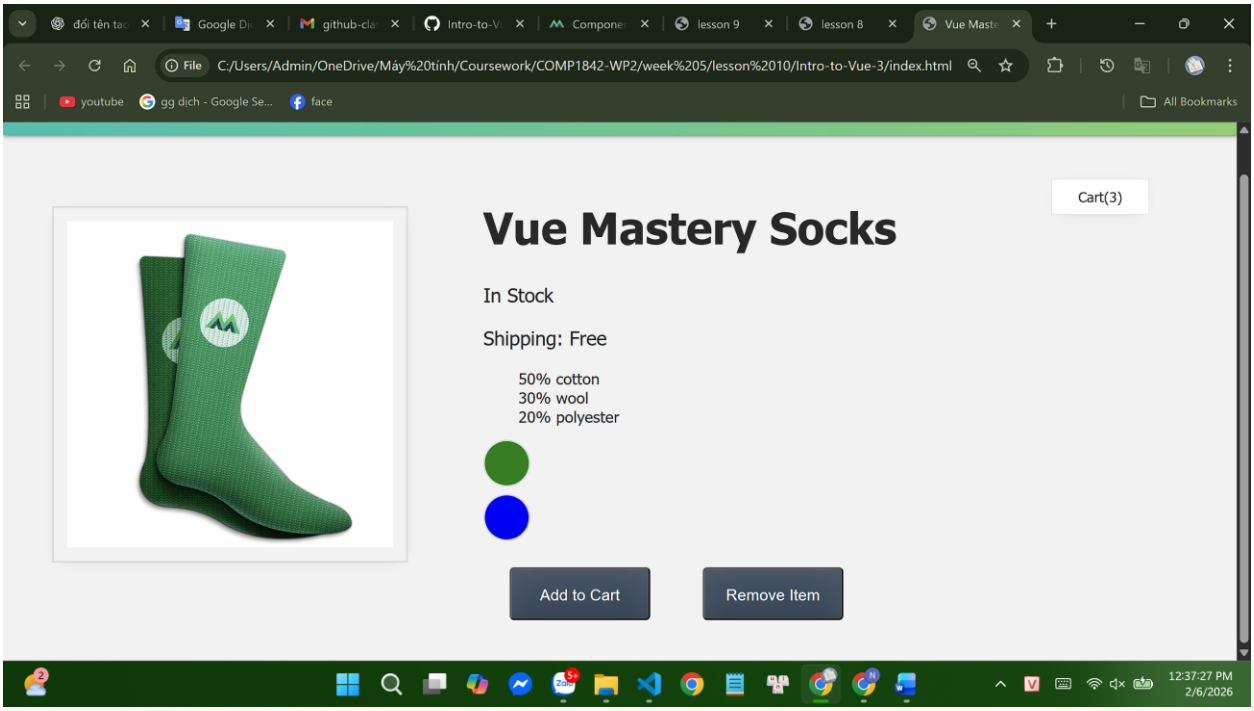
```javascript
//Ngo Quoc Cuong
const app = Vue.createApp({
    data() {
        //exercise
        return {
            cart: [],
            premium: true
        }
        //exercise
    },
    methods: {
        //exercise
        updateCart(id) {
            this.cart.push(id)
        },
        //exercise
        //challenge
        removeById(id) {
            const index = this.cart.indexOf(id)
                if (index > -1) {
                    this.cart.splice(index, 1)
                }
        }
        //challenge
    }
})
```

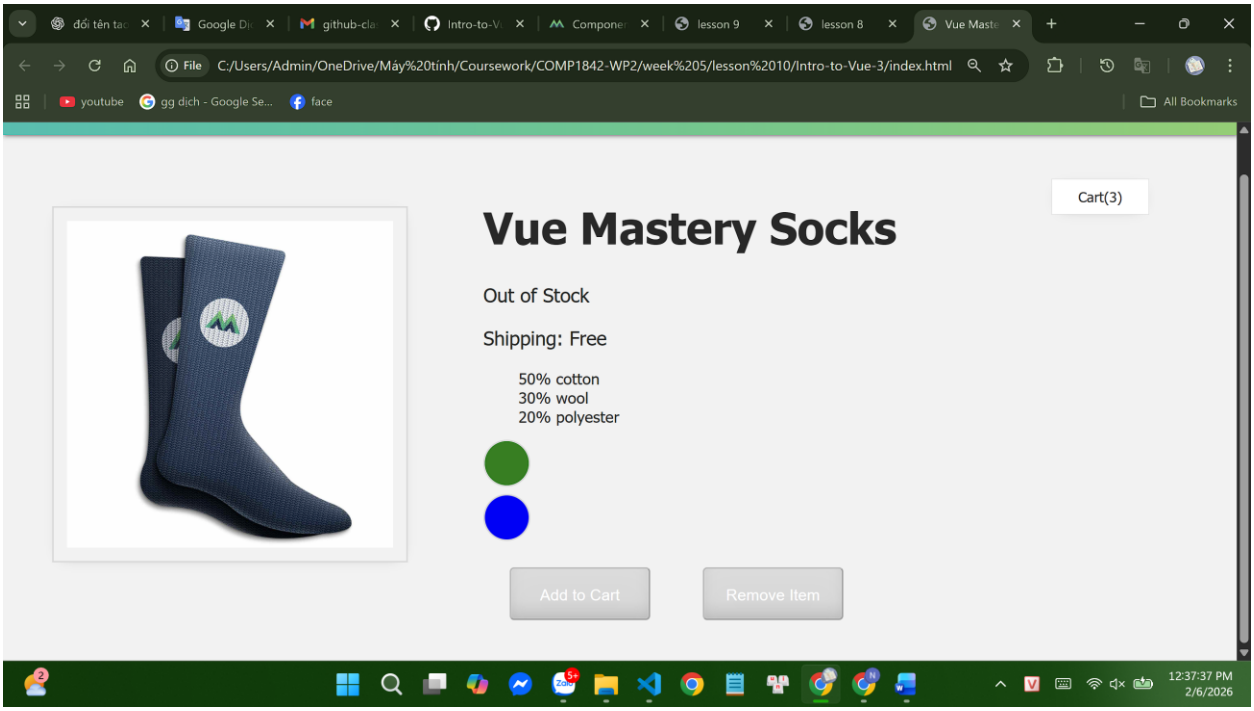
```javascript
app.component('product-display', {
  props: {
    premium: {
      type: Boolean,
      required: true
    }
  },
  template:
  /*html*/
  `<div class="product-display">
    <div class="product-container">
      <div class="product-image">
        <img v-bind:src="image">
      </div>
      <div class="product-info">
        <h1>{{ title }}</h1>

        <p v-if="inStock">In Stock</p>
        <p v-else>Out of Stock</p>

        <p>Shipping: {{ shipping }}</p>
        <ul>
          <li v-for="detail in details">{{ detail }}</li>
        </ul>

        <div
          v-for="(variant, index) in variants"
          :key="variant.id"
          @mouseover="updateVariant(index)"
```

```javascript
  template:
          @mouseover="updateVariant(index)"
          class="color-circle"
          :style="{ backgroundColor: variant.color }">
        </div>

        <button
          class="button"
          :class="{ disabledButton: !inStock }"
          :disabled="!inStock"
          v-on:click="addToCart">
          Add to Cart
        </button>

        <!--challenge-->
        <button
        class="button"
        :class="{ disabledButton: !inStock }"
        :disabled="!inStock"
        @click="removeFromCart">
        Remove Item
        </button>
        <!--challenge-->

        </div>
      </div>
    </div>`,
  data() {
    return {
```

```javascript
    data() {
      return {
        product: 'Socks',
        brand: 'Vue Mastery',
        selectedVariant: 0,
        details: ['50% cotton', '30% wool', '20% polyester'],
        variants: [
          { id: 2234, color: 'green', image: './assets/images/socks_gre
          { id: 2235, color: 'blue', image: './assets/images/socks_blue
        ]
      }
    },
    methods: {
      //exercise
      addToCart() {
        this.$emit('add-to-cart', this.variants[this.selectedVariant]
      },
      //exercise

      //challenge
      removeFromCart() {
        this.$emit('remove-from-cart', this.variants[this.selectedVaria
      },
      //challenge
      updateVariant(index) {
        this.selectedVariant = index
      }
    },
    computed: {
```

```javascript
  computed: {
      title() {
        return this.brand + ' ' + this.product
      },
      image() {
        return this.variants[this.selectedVariant].image
      },
      inStock() {
        return this.variants[this.selectedVariant].quantity
      },
      shipping() {
        if (this.premium) {
          return 'Free'
        }
        return 2.99
      }
    }
})
```





4. Understanding

After completing lessons 8 through 10 in the Vue 3 Introduction course on Vue Mastery, I gained a clearer and more practical understanding of how Vue helps developers build well-structured applications. In lesson 8, I learned about computed properties and how they are used to compute values based on existing data. Instead of directly placing calculations in the HTML template, computed properties allow us to write logic in a

single place. Vue automatically updates the results when the relevant data changes, making the code easier to understand and more efficient. Lesson 9 focuses on components, one of the most important concepts in Vue. Components allow us to divide the user interface into smaller parts, such as headers, product sections, or buttons. Each component can manage its own data and logic. This helps keep the project well-organized and makes the code easily reusable across different parts of the application. Finally, in Lesson 10, I learned about props, which are used to pass data from a parent component to a child component. Props allow the same component to display different data without changing its structure. I also learned that props should not be modified inside a child component, which helps prevent unexpected issues. Overall, these lessons helped me understand how Vue manages data, structure, and communication between components in a simple and clear way.